Research Statement

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I am an algebraic topologist and my work explores the connections between algebraic topology, number theory, and geometric topology. My research program is primarily in two areas: the intersection of algebraic K-theory and chromatic homotopy theory and the intersection of equivariant topology and Hochschild homology. My work in algebraic K-theory and chromatic homotopy theory is primarily aimed towards understanding the arithmetic of ring spectra, which generalize rings. My program on equivariant topology and Hocschild homology has potential applications to geometric topology that I aim to explore in future work. The common thread in my research is the study of invariants of ring spectra.

One of the main families of conjectures in the area of periodicity and algebraic K-theory are the Ausoni–Rognes red-shift conjectures, which generalize famous conjectures of Lichtenbaum and Quillen. My work gives quantitative evidence for the red-shift conjectures from several different perspectives. I have done this by developing tools for computing Hochschild homology [AK18], giving new computations of Hochschild homology using these tools [AK17, AK21, ACH21, AAC$^+$22], and building on these computations to give evidence [AK18, AS20, AQ19, AK22] for the Ausoni–Rognes red-shift conjectures in algebraic K-theory. I also have several ongoing projects extending this evidence and computational knowledge [AHW, ACH, AAHR]. These results are summarized in Section 1.

Equivariant topology has had a renaissance recently due to groundbreaking work of Hill, Hopkins, and Ravenel on the Arf–Kervaire invariant problem in geometric topology. One of my long-term goals is to resolve the 3-primary Arf–Kervaire invariant problem using an deformation of equivariant topology. My work [AQ21] uses equivariant topology and Hochschild homology to give a first step towards computing the stable h-cobordism space of a point. My work [AGH21] gives new perspectives on Hochschild homology and Witt vectors of rings with anti-involution. My work in progress [AMP], gives a combinatorial approach to equivariant Hochschild homology. One goal of this program is to study equivariant knot Floer homology using this theory. In work in progress [ABBK], we also use deformations of homotopy theory to study C_3-equivariant homotopy. Finally, in work in progress [AQ] we study a motivic filtration on Hochschild homology of rings with anti-involution. These results are summarized in Section 2.1.

1 Algebraic K-theory and chromatic homotopy

This section consists of two subsections: Section 1.1 includes a brief survey of chromatic homotopy theory aimed at a broad audience in Section 1.1.1 and a brief survey of the trace methods approach to computing algebraic K-theory in Section 1.1.2. Section 1.2 contains a selection of my results on the arithmetic of ring spectra as well as future research directions.

1.1 Background

1.1.1 Chromatic homotopy theory

One of the fundamental open problems in algebraic topology is the computation of homotopy groups of spheres. These groups are important because we want to classify CW complexes,

which are topological spaces built out of spheres. The homotopy groups of spheres are the the gluing data for assembling these nice spaces. In the 1930's, H. Freudenthal [Fre38] showed that the computation of the homotopy groups of spheres stabilize in high dimensions and this was a starting point of stable homotopy theory. Many problems in geometric topology, such as the study of bordism groups, are naturally stable phenomena.

The stable homotopy groups are much simpler and more accessible via successive algebraic approximations known as spectral sequences. One of our favourite spectral sequences is a spectral sequence constructed by J. F. Adams in the 1950's [Ada58], henceforth called the Adams spectral sequence. A variant of this spectral sequence, known as the Adams–Novikov spectral sequence was later constructed whose input is the cohomology of a Hopf algebroid associated to complex cobordism MU. The field of chromatic homotopy theory is rooted in work of Quillen [Qui69] from the 1960's relating this Hopf algebroid to the theory of formal groups. Formal groups can be classified by their height in the sense that there is a height stratification of the p-localization of the moduli stack of formal groups

$$\cdots \subset \mathcal{M}_{FG,(p)}^{\leq n} \subset \mathcal{M}_{FG,(p)}^{\leq n+1} \subset \cdots \subset \mathcal{M}_{FG,(p)}$$

by open substacks. The tight connection between the Formal groups and complex cobordism can be combined with the Adams–Novikov spectral sequence to transfer this height stratification into the world of spectra. In the 1970's, H. Miller, D. Ravenel, and W. S. Wilson [MRW75] pioneered work on the chromatic spectral sequence, which computes the cohomology of the Hopf algebroid associated to complex cobordism and is closely tied to the height stratification of the moduli stack of formal groups. In the input of the chromatic spectral sequences there are elements known as the Greek letter family elements depending on a non-negative integer n and a prime p, corresponding to the moduli stack of p-local formal groups of height $\leq n$. For small enough n and large enough p these elements are known to produce families of elements in the stable homotopy groups of spheres known as the α-family, β-family, and γ-family corresponding to heights one, two, and three respectively. When viewed through this lens, the homotopy groups of spheres consist of elements of longer and longer wavelength and, by analogy with light spectrometry, we think of these wave lengths as corresponding to the increasing wavelengths of light in the color spectrum. This is where the term chromatic comes from in chromatic homotopy theory.

As the height increases the light moves in the red direction by the analogy with light spectometry. This is why conjectures of C. Ausoni and J. Rognes, which are an important part of my research program, are called the red-shift conjectures. These conjectures are part of a overarching philosophy of how invariants of associative ring spectra such as algebraic K-theory, topological cyclic homology, and topological periodic cyclic homology, increase chromatic complexity as exhibited by explicitly increasing the wavelength of these periodic classes. Though this phenomena is tantalizing to a stable homotopy theorist, the more exciting aspect of these conjectures is that they give a precise formulation of how one may study the arithmetic of ring spectra and this is my primary motivation.

1.1.2 Trace methods

In the 1970's, Quillen [Qui73] first constructed a spectrum $K(R)$ associated to a ring R whose homotopy groups are the algebraic K-theory groups $K_*(R)$ or the ring. This construction takes linear algebraic data as input and miraculously produces deep arithmetic

data as output. Since algebraic K-theory is defined quite broadly, we may also consider the algebraic K-theory of an associative ring spectrum R and use algebraic K-theory to study the arithmetic of R. Unfortunately for us, algebraic K-theory is notoriously difficult to compute, however in the 1980's a new approach to computing algebraic K-theory was developed by M. Bökstedt and the first example of its power was demonstrated in groundbreaking work of M. Bökstedt, W. Hsiang, and I. Madsen [BHM93]. In 2018, T. Nikolaus and P. Scholze [NS18] significantly simplified the story, as I discuss below and this has lead to flood of new results in this area.

In the theory of trace methods, one computes algebraic K-theory by successive approximations. The first approximation to algebraic K-theory is Hochschild homology. Classically, one uses genuine equivariant homotopy theory and genuine cyclotomic structure to construct $\mathrm{TR}(R)$ from Hochschild homology and then define p-complete TC as the homotopy equalizer

$$\mathrm{TC}(R)^\wedge_p = \mathrm{eq}\left(\mathrm{TR}(R) \underset{F}{\overset{\mathrm{id}}{\rightrightarrows}} \mathrm{TR}(R) \right)$$

of the identiy map and a map F, originally called the Frobenius map because of the relation to the theory of Witt vectors. In particular, the groups $\mathrm{TR}_*(R)$ are interesting in their own right because they recover the de Rham–Witt complex [HM03].

The Dundas–Goodwillie–McCarthy theorem [DGM13] then proves that topological cyclic homology is a close approximation to algebraic K-theory. In particular, if $f\colon A \to B$ is an infinitesimal extension and we let $E(A,B)$ denote the fiber of the map $E(f)$ for $E \in \{K, \mathrm{TC}\}$, then there is an equivalence $\mathrm{K}(A,B) \simeq \mathrm{TC}(A,B)$ between relative algebraic K-theory and relative topological cyclic homology. As a consequence, for nice enough ring spectra algebraic K-theory and topological cyclic homology are equivalent after completion.

Groundbreaking work of Nikolaus–Scholze [NS18] simplified the computational approach to topological cyclic homology $\mathrm{TC}(R)$ by describing TC as the homotopy equalizer

$$\mathrm{TC}(R) = \mathrm{eq}\left(\mathrm{TC}^-(R) \underset{\varphi}{\overset{can}{\rightrightarrows}} \mathrm{TP}(R)^\wedge \right)$$

of two maps where $(-)^\wedge$ denotes pro-finite completion. The map can comes from an action of the circle S^1 on Hochschild homology and the map φ comes from the cyclotomic structure on Hochschild homology. The topological negative cyclic homology $\mathrm{TC}^-_*(R)$ is the S^1-homotopy fixed points of Hochschild homology and the topological periodic cyclic homology $\mathrm{TP}_*(R)$ is the S^1-Tate construction of Hochschild homology. These invariants are topological analogues of (algebraic) negative cyclic homology and (algebraic) periodic cyclic homology respectively. In contrast to TR, these invariants are potentially more computable using the homotopy fixed point and Tate spectral sequence respectively. These intermediate approximations $\mathrm{TC}^-(R)$ are $\mathrm{TP}(R)$ to algebraic K-theory theory are interesting in their own right as well because of the relation to prismatic cohomology [BS19]. This gives an explicit computational approach to algebraic K-theory, which has led to numerous advances in the subject, for example see [HM03, AR02, HW20, AAC+22, HRW22] for selection of results of particular relevance here.

1.2 Results

As exhibited at the end of Section 1.1.1, the Ausoni–Rognes red-shift conjectures are broadly about the arithmetic of ring spectra. In particular, these conjectures generalize famous conjectures of S. Lichtenbaum [Lic73] and D. Quillen [Qui75] from the 1970's. Many of the conjectures of S. Lichtenbaum and D. Quillen have since been resolved by work of M. Rost and V. Voevodsky [Voe11]. A qualitative version of the red-shift philosphy of Ausoni–Rognes was resolved in 2022 for all commutative ring spectra by impressive work of Burkland–Schlank–Yuan [BSY22]. However, the red-shift conjectures as originally stated explicitly by C. Ausoni and J. Rognes in [AR08] are still open as well as a stronger version originally posed by J. Rognes [Rog00]. Here, I also pose my own question, which I view as a higher chromatic height analogue of a conjecture of Lichtenbaum [Lic73] motivated by my work on the iterated algebraic K-theory of the integers.

1.2.1 Higher Lichtenbaum

In the late 1960's, Adams [Ada66] and Quillen [Qui72] showed a height one periodic family known as the α-family is detected in $\mathrm{K}_{4k-1}(\mathbb{Z})$. This the first evidence for a conjecture of Lichtenbaum [Lic73, Conjecture 2.4] that certain quotients of orders of algebraic K-theory groups

$$|\,\mathrm{K}_{4k+2}(\mathcal{O}_F)|/|\,\mathrm{K}_{4k+3}(\mathcal{O}_F)| = \zeta_F(-2k-1)$$

can be identified (up to a sign and power of two) with special values of the Dedekind zeta function ζ_F of a (totally real) number field F with ring of integers $\mathcal{O}_F$. My work [AK18] gives the first indication of a higher chromatic height analogue of Lichtenbaum's conjecture.

Question 1.1. [Higher Lichtenbaum] Is there a precise relation between $\pi_* K(K(\mathcal{O}_F))$ and modular forms over $\mathcal{O}_F$?

Just as detecting the α-family was the first evidence [Lic73, §3] for Lichtenbaum's conjecture, we consider the detection of the β-family in iterated algebraic K-theory of the integers as a first step towards Question **??**. Recall from Section 1.1.1 that there is a height two periodic family of elements in the p-local stable homotopy groups of spheres known as the β-family for large enough primes. In [Lau99, Beh06, BL09], M. Behrens and G. Laures demonstrated a tight connection between the p-primary β-family for $p \geq 5$ and integral modular forms. The following result may therefore be viewed as a higher chromatic height analogue of the results of Adams [Ada66] and Quillen [Qui72] motivating [Lic73, Conjecture 2.4] and gives evidence for Question 1.1.

Theorem 1.2 ([AK18])**.** The p-primary β-family modulo (p, v_1) is detected in the iterated algebraic K-theory $\pi_* \mathrm{K}(\mathrm{K}(\mathbb{Z}))$ of the integers for $p \geq 5$.

1.2.2 Ausoni–Rognes red-shift conjecture

The result [AK18] built on computations in my PhD thesis [AK17], published in [AK21] and joint work with Salch [AS18]. In [AS18], we provided a useful tool for computing Hochschild homology of commutative ring spectra. For simplicity, we state the result only for Hochshild

homology (THH) though it works also for generalizations such as factorization homology. By a multiplicatively filtered commutative ring spectrum, I mean a sequence of spectra

$$\longrightarrow I_2 \longrightarrow I_1 \longrightarrow I_0$$

equipped with structure maps $\rho_{i,j}\colon I_i \wedge I_j \longrightarrow I_{i+j}$ for all $i, j \geq 0$ satisfying commutativity, associativtiy, and unitality conditions. Let E_0^* denote the associated graded functor from multiplicatively filtered commutative ring spectra to commutative ring spectra.

Theorem 1.3 ([AS18])**.** Let I be a multiplicatively filtered commutative ring spectrum, then there is a spectral sequence

$$\mathrm{THH}_*(E_0^* I) \implies \mathrm{THH}_*(I_0)$$

for any generalized homology theory F.

In [AK21], I used this spectral sequence to compute mod (p, v_1) Hochschild homology of the connective cover of the $K(1)$-local sphere, denoted simply j, at primes $p \geq 5$. This was also computed by [Hön21] using different methods.

Theorem 1.4 ([AK21])**.** Let $p \geq 5$. There is an isomorphism of graded $\mathbb{F}_p$-algebras

$$V(1)_* \mathrm{THH}(j) \cong H(E(\alpha_1, \lambda_1, \lambda_2); d) \otimes P(\mu_2) \otimes \Gamma(\sigma b)$$

where $d(\lambda_2) = \alpha_1 \lambda_1'$.

Since j is the connective cover of the $K(1)$-local sphere and one could use a localization sequences to compute the algebraic K-theory of $L_{K(1)}\mathbb{S}$ from j, this also gives some of progress on one of the most impenetrable Ausoni–Rognes red-shift conjectures. To phrase the conjecture, note that there exists a finite p-local spectrum $F(n)$ of type n, which implies that there exists a self map v inducing an isomorphism in $K(n)_*$ whose telescope $v^{-1}F(n)$ we denote $T(n)$. We say that X has telescopic complexity n if the fiber of the canonical map $F(n) \otimes X \to T(n) \otimes X$ is bounded above. The following conjecture appears in [AR08, Conjecture 4.3].

Conjecture 1.5 (Ausoni–Rognes red-shift conjecture)**.** Does $K(L_{K(n)}\mathbb{S})$ have telescopic complexity $n + 1$?

1.2.3 Qualitative red-shift

The Ausoni–Rognes red-shift conjecture 1.5 remains unresolved, but a more tractable conjecture was posed for commutative ring spectra by L. Meier, M. Land, and G. Tamme. This conjecture qualitatively captures the red-shift philosphy of Ausoni–Rognes.

Question 1.6 (Qualitative red-shift)**.** For a ring spectrum R then is $K(m)_*R \neq 0$ if and only if $K(m+1)_*R \neq 0$?

When R is a commutative ring spectrum than there exists an integer n such that $K(m)_*R = 0$ for all $m \leq n$ and $K(m)_*R = 0$ for all $m > n$ by work of J. Hahn [Hah16]. We may therefore say that a commutative ring spectrum has height n if $K(n)_*R \neq 0$ and $K(n+1)_*R = 0$. For commutative ring spectra, Question 1.6 has now been proven by a tour de force beginning with work of [CMNN20] and [LMMT20] and culminating in results of [Yua21] and [BSY22]. We therefore know that the red-shift philosohpy is qualitatively true for all commutative ring spectra.

It is also known that Question 1.6 is too much to ask for associative ring spectra, for example in [AHW] we show that $K(n)_*K(K(n)) \neq 0$ even though $K(n-1)_*K(n) = 0$. Nonetheless, it is interesting to consider versions of Question 1.6 for other approximations to algebraic K-theory such as topological periodic cyclic homology. Question 1.6 is also open for E_m ring spectra which are not commutative ring spectra, though [CMNN20] and [LMMT20] have proven a vanishing result in this case too and [HW20] have proven that this for truncated Brown–Peterson spectra, which are known to be E_3 ring spectra by their work and also known to not be E_∞-ring spectra by [Law18, Sen17].

My own work on this question slightly predates these other recent results. For my first result, note that there is a family of spectra $y(n)$ depending on a non-negative integer n and prime p that interpolate between $\mathbb{S}$ and $H\mathbb{F}_2$

$$\mathbb{S} = y(0) \longrightarrow y(1) \longrightarrow \cdots \longrightarrow y(\infty) = H\mathbb{F}_2.$$

These spectra were originally constructed by M. Mahowald [Mah79] and M. Mahowald, D. Ravenel, and P. Schick [MRS01] in their attempted disproof of the telescope conjecture. These spectra are interesting from the perspective of chromatic homotopy theory because $K(m)_*y(n) \neq 0$ for $0 \leq m < n$ so they behave like the type n spectra $F(n)$ introduced earlier even though they are not finite spectra. We prove a vanishing version of Question 1.6 for $\mathrm{TP}(y(n))$.

Theorem 1.7 ([AQ19])**.** The spectrum $\mathrm{TP}(y(n))$ is $K(m)$-acyclic for $1 \leq m \leq n$. Also, $K(y(n))$ is $K(m)$ acyclic for $1 \leq m \leq n-1$.

The fact that $K(y(n))$ is $K(m)$ acyclic for $1 \leq m \leq n-1$ was also proven later as a consequence of [CMNN20]. This result uses a technical result about commuting Morava K-theory with certain sequential limits, whihc I proved with A. Salch [AS20]. As an application of this technical result, we also proved an upper Morava K-theory vanishing region for commutative ring spectrum forms of $BP\langle n\rangle$. Note that $L_{K(m)}BP\langle n\rangle = 0$ for $m \geq n+1$.

Theorem 1.8 ([AS20])**.** When B is a commutative ring spectrum form of $BP\langle n\rangle$, then $L_{K(m)}K(B) = 0$ for $m \geq n+2$.

Soon after we posted our preprint, a more general result was proven by [LMMT20] using entirely complementary methods. Our work [AS20] is different from [LMMT20] in that it also sheds light on TC^- and TP providing additional applications as exemplified by [AQ19].

1.2.4 Pure fp-type red-shift conjecture

A strong form of the red-shift conjectures was originally posed by J. Rognes in [Rog00]. I will refer to it as the pure fp-type red-shift conjecture below. To phrase the conjecture, we

need a definition. Note that since a finite type n spectrum $F(n)$ has a periodic self map v, the homotopy groups $F(n)_*X$ are always a $\mathbb{F}_p[v]$-module. We say that a spectrum has pure fp-type n if the $\mathbb{F}_p[v]$-module $F(n)_*(X)$ is a finitely generated, free module following J. Rognes [Rog00] based on a definition of M. Mahowald and C. Rezk [MR99]. If X has pure fp-type n then X has fp-type n in the sense of [MR99] and it also has telescopic complexity n. A commutative ring spectrum with telescopic complexity n also has height n. Therefore Question 1.9 is a stronger statement than Conjecture 1.5, which is also a stronger statement than Question 1.6.

Question 1.9 (Pure fp-type red-shift conjecture)**.** If R is a ring spectrum with pure fp-type n, then does $\mathrm{TC}(R)$ have pure fp-type $n+1$?

This was posed by J. Rognes based on computations of the homotopy of $V(1)_*K(\ell)$ where $V(1) = \mathbb{S}/(p, v_1)$ is a four cell complex which has type 1. In particular, they proved that it is a finitely generated free $\mathbb{F}_p[v_2]$-module and therefore Conjecture 1.9 holds in this case. To do this, they begin by computing Hochschild homology (THH).

The truncated Brown–Peterson spectra are ring spectra of pure fp-type n, which are fundamental in chromatic homotopy theory. For small n these truncated Brown–Peterson spectra have more familiar names: by convention $BP\langle -1\rangle = H\mathbb{F}_p$, we have $BP\langle 0\rangle = H\mathbb{Z}_{(p)}$, and $BP\langle 1\rangle = \ell$, which is the Adams summand of p-local complex K-theory ku. The spectrum $BP\langle 2\rangle$ can also be constructed using versions topological modular forms and topological automorphic forms [HL10, LN12].

The calculations of $\mathrm{THH}_*(BP\langle n\rangle)$ for $n \geq -1$ are deep results and they are increasingly difficult as n increases. For example, Bökstedt's result $\mathrm{THH}_*(BP\langle -1\rangle) = P(\mu_0)$ [Bök87] can be used to prove Bott periodicity [HN19]. When describing the torsion in $\mathrm{THH}_*(BP\langle 1\rangle)$, Angeltveit–Hill–Lawson say it "follows a kind of tower-of-Hanoi pattern with increasingly complicated, inductively built, components" [AHL10, p.6]. In future work, I plan to give a complete description of $\mathrm{THH}_*(BP\langle 2\rangle)$. Towards this aim, I have already computed topological Hochschild homology with various coefficients in joint work with D. Culver and E. Höning.

Theorem 1.10 ([ACH21])**.** When $m = 0, 1, 2$, there are isomorphisms

$$\mathrm{THH}_*(BP\langle 2\rangle; k(m)) \cong P(v_m) \otimes E(\lambda_1, \ldots, \lambda_{2-m}) \otimes T^n_m,$$

where T^n_m is an explicit v_m-torsion module.

In work in progress, I use different filtration techniques such as Theorem 1.3 and Bockstein spectral sequences to compute $\mathrm{THH}_*(BP\langle 2\rangle; \ell)$ in [ACH].

In joint work [AAC+22] with C. Ausoni, D. Culver, E. Höning, and J. Rognes I compute $V(2)_*\,\mathrm{TC}(BP\langle 2\rangle)$ and $V(2)_*\,\mathrm{K}(BP\langle 2\rangle)$ at primes $p \geq 7$ improving on a result from the landmark paper of Hahn–Wilson [HW20] in the case $n = 2$. Hahn–Wilson [HW20] prove that $F(n)_*\,\mathrm{TC}(BP\langle n\rangle)/(v_n)$ is finite, or in other words $\mathrm{TC}(BP\langle n\rangle)$ has fp-type $n+1$. This also implies that algebraic K-theory of $BP\langle n\rangle$ has fp-type $n+1$ and telescopic complexity $n+1$. Our result is currently the first and only known proof of Conjecture 1.9 at higher heights than the original result of C. Ausoni and J. Rognes [AR02]. Note that $BP\langle 2\rangle$ has pure fp-type 2.

Theorem 1.11 ([AAC+22])**.** The spectrum $\mathrm{TC}(BP\langle 2\rangle)$ has pure fp-type 3.

1.2.5 Syntomic cohomology of $k(n)$

Recent work of Hahn–Raksit–Wilson [HRW22] makes computations of algebraic K-theory and topological cyclic homology more accessible. In op. cit., the authors provide a motivic filtration on topological cyclic homology of commutative ring spectra, which is amenable to computation for certain ring spectra called chromatically quasi-syntomic ring spectra. As an application they extend the computation of $V(1)_* \operatorname{TC}(BP\langle 1\rangle)$ to $p = 3$ and provide a much simpler approach to the computation for all primes $p \geq 3$. Even though ring spectra such as the truncated Brown–Peterson spectra $BP\langle n\rangle$ and the connective cover of Morava K-theory $k(n)$ are not strictly speaking chromatically quasisyntomic, they are approachable by similar means. This is the subject of ongoing work with J. Hahn and D. Wilson.

For chromatically quasisyntomic ring, the motivic filtrations on TC and TP due to [HRW22] extend the motivic filtration on TC and TP by [BMS19, BS19, BL22], whose associated graded are syntomic cohomology and (absolute) prismatic cohomology, respectively. One may therefore define syntomic cohomology and prismatic cohomology for ring spectra using this approach. I will take this as a definition of syntomic cohomology and prismatic cohomology for an arbitrary ring spectrum.

Goal 1.12. Compute the syntomic cohomology of truncated Brown–Peterson spectra $BP\langle n\rangle$, connective Morava K-theory $k(n)$.

The syntomic cohomology is the E_2-page of a spectral sequence computing $\operatorname{TC}(R)$ and it often collapses at this page for bidegree reasons. For example, this is the case for $V(1)_*K(BP\langle 1\rangle)$ by [HRW22]. In general, there may be longer differentials in this spectral sequence. Nevertheless, there are questions that you can answer using this approach. We say that X has fp-type n if there exists a finite spectrum of type $n+1$ such that $F(n+1)_*X$ is finite. The following conjecture is a slightly weaker version of the question of Rognes [**?**].

Question 1.13. If R is a ring spectrum of fp-type n, then does $K(R)$ have fp-type $n+1$?

Note that a spectrum whose homology is finitely presented as a comodule over the dual Steenrod algebra is telescopic complexiy n if and only if it has fp-type n, so Question 1.13 is Conjecture 1.5 applied more generally.

Goal 1.14. Prove that $K(k(n))$ has fp-type $n+1$.

1.2.6 Algebraic K-theory of the image of J

Even though j is a commutative ring spectrum, since MU_*j is not even it is not a chromatically quasisyntomic ring. However, it is equipped with a cosimplicial resolution by even commutative ring spectra and one may use this to approach the algebraic K-theory of the image of J. Note that j has pure fp-type 1, but it is not a regular ring spectrum. One expects that a regularity hypothesis is needed for Question 1.13 to hold.

Goal 1.15. Determine if $\operatorname{TC}(j)$ has fp-type 2.

Recall that $T(n)$ denotes the telescope $v_n^{-1}F(n)$ of a finite type n spectrum. The telescope conjecture states that $L_n = L_n^f$ where L_n is the localization at $K(0) \oplus \cdots \oplus K(n)$ and L_n^f

is the localization at $T(0) \oplus \cdots \oplus T(n)$. It is now widely believed to be false. In [MR99], they conjecture that the telescope conjecture is true for all fp-spectra, so if one believes this conjecture than an spectrum that is not fp (does not have finiltely presented mod p cohomology as a module over the Steenrod algebra) is a good choice for studying the red-shift conjecture. Explicit computations also suggest that $V(1)_*K(j))$ may be a good test case for the telescope conjecture, as has been suggested by J. Hahn. We say that the telescope conjecture holds for X if $L_nX = L_n^fX$. Note that if there are any such X then the telescope conjecture is false, which would resolve the only outstanding conjecture from a famous family of conjectures known as the red-shift conjectures.

Goal 1.16. Determine whether the telescope conjecture holds for $V(1) \otimes K(j)$.

1.2.7 G-theory

In 2012, C. Ausoni and J. Rognes [AR12] computed the algebraic K-theory of the first connective Morava K-theory $k(1)$ modulo (p, v_1). As part of this program, they proposed that one should consider localization sequences

$$K(L/p) \to K(L_p^\wedge) \to K(ffL_p^\wedge) \tag{1}$$

where the third term is defined as the cofiber of the transfer map associated to the map of ring spectra $i\colon L_p^\wedge \to L/p$ [Rog14, Definition 6.1]. This is in analogy with the classical localization sequence

$$K(\mathbb{F}_p) \to K(\mathbb{Z}_p^\wedge) \to K(\mathbb{Q}_p)$$

of Quillen [Qui73]. In particular, Rognes [Rog14] suggests that there exists a spectral sequence

$$H^{-s}_{\mathrm{mot}}(ffL_p^\wedge; \mathbb{F}_{p^2}(t/2)) \implies V(1)_*K(ffL_p^\wedge)$$

computing the algebraic K-theory of this cofiber modulo (p, v_1).

One may consider similar questions for other elements $v_m \in \pi_*BP\langle n\rangle$, for example B. Antieau, T. Barthels, and D. Gepner [ABG18b] show that there is a localization sequence

$$K(\mathrm{End}_{BP\langle n\rangle}(BP\langle n-1\rangle)) \to K(BP\langle n\rangle) \to K(BP\langle n\rangle[v_n^{-1}]).$$

When $n = 1$, this has an even nicer form by earlier work of Blumberg–Mandell [BM08]

$$K(H\mathbb{Z}) \to K(BP\langle 1\rangle) \to K(BP\langle 1\rangle[v_1^{-1}])$$

which compares nicely with the localization sequence above, but this fails for larger n. Nonetheless, these localization sequences were used to show that $K(m)_*K(BP\langle n\rangle[v_n^{-1}]) = 0$ for $m \geq n+2$ [LMMT20, Corollary D].

Algebraically, there are two different notions of algebraic K-theory commonly considered known as K-theory and G-theory. K-theory is defined using finitely generated projective modules where as G-theory is defined using pseudo-coherent modules. For regular Noetherian rings, these two notions agree.

For ring spectra, we still do not have a good notion of G-theory even though algebraic K-theory is defined quite broadly. One reason is that the right notion of pseudo-coherent

modules has not been developed. Another other wrinkle in this story is that the algebraic K-theory of the fraction field $ffL_p^\wedge$ may not actually be the algebraic K-theory of a ring spectrum, but rather a ring spectrum with many objects. This makes the theory a bit more complicated. Nonetheless, we soldier on.

In joint work with C. Ausoni, E. Höning, and J. Rognes, we are working on developing the theory of G-theory for ring spectra equipped with localization sequences such as (1). In fact, one needs to develop this theory in a way that is flexible enough to contain ring spectra with many objects. Ultimately, one may ambitiously hope to realize the algebraic K-theory of a certain maximal Galois extension of ℓ as a close approximation to Lubin–Tate E-theory E_2 as conjectured in [AR08, Conjecture 4.4].

2 Equivariant and motivic topology

This section is split into two sub-sections: Section 1.1 gives some background on equivariant topology and motivic topology and Section 2.2 discusses results and work in progress.

2.1 Background

2.1.1 Equivariant algebra and topology

The field of equivariant topology was revitalized in the last decade with the groundbreaking work of Hill, Hopkins, and Ravenel [HHR16] resolving almost all remaining cases of the Arf–Kervaire invariant one problem. Since this work, there has been significant advances in the subject building on the machinery in their paper. One of the foundational parts of their work is the norm functor

$$N_H^G\colon \mathsf{Sp}^H \longrightarrow \mathsf{Sp}^G$$

from genuine H-spectra to genuine G-spectra, which they define for any finite group G.

The work of [HHR16] also relies on the theory of genuine commutative G-ring spectra. One may ask what the associative analogue of a genuine commutative G-ring spectrum is and one answer, in the case when G is the cyclic group of order 2, is a ring spectrum with anti-involution. Ring spectra with anti-involution play a key role in surgery theory as they are the input for Real algebraic K-theory, a genuine equivariant version of algebraic K-theory, which recovers various forms of L-theory. Real algebraic K-theory also recovers various forms of Grothendieck–Witt theory and it is therefore interesting from an algebraic and algebro-geometric perspective as well. Just as classically algebraic K-theory can be approximated by topological Hochschild homology, Real algebraic K-theory can be approximated by Real topological Hochschild homology [HNS22]. One may then form Real analogues of topological cyclic homology and related invariants [QS21] and use them to approximate Real algebraic K-theory.

2.1.2 Motivic topology

In motivic topology, one replaces topological spaces with schemes and then does homotopy theory in this setting. The role of the interval, which is contractible classically, is played by the affine line $\mathbb{A}^1$. One therefore considers pre-sheaves on the category of smooth schemes

over a base scheme $X = \mathrm{spec}(R)$ for some commutative ring R, localizes in the Nisnevich topology, and then inverts $\mathbb{A}^1$ so that the category behaves more like the category of topological spaces. Motivically, the scheme $\mathbb{G}_m$ behaves like a second sphere and smashing this with the usual sphere produces projective space $\mathbb{P}^1$. The analogue of infinite loops spaces are then infinite loop spaces with respect to the sphere $\mathbb{P}^1$ and stabilizing with respect to $\mathbb{P}^1$ produces the stable category $\mathrm{SH}(R)$.

The motivic stable category subcategory of cellular complex motivic spectra $\mathrm{SH}^{\mathrm{A}}(\mathbb{C})$ is equipped with a realization functor

$$\mathrm{Be}\colon\ \mathrm{SH}^T(\mathbb{C}) \to \mathsf{Sp}$$

to the category of spectra, called Beti realization. This functor arises by taking the complex points of scheme X over $\mathrm{spec}(\mathbb{C})$. Based on work of [GIKR18, Pst18], this may be viewed as deformation of spectra. Moreover, the extra grading on the homotopy groups in $\mathrm{SH}^T(\mathbb{C})$ coming from the additional spheres $\mathbb{G}_m^{\otimes t}$ has led to significant progress on computations, such as the homotopy groups of spheres by Isaksen–Wang–Xu [IWX20]. In fact, any time one has a deformation of a stable ∞-category one may use similar techniques to do computations and this has led to numerous advances in the subject, for example [BHS19].

2.2 Results

2.2.1 Real topological Hochschild homology

Even though [HHR16] define norms for finite groups, it remains open to define norms more generally for compact Lie groups in general and it is under active investigation. So far it has only been done in certain cases. In [ABG+18a], they define norms $N_{\mu_n}^{\mathbb{T}}(R)$ from the groups of unity to the circle group $\mathbb{T}$. In fact, this is done by showing that the cyclic bar construction has the universal property of the norm. In joint work with Gerhardt and Hill [AGH21], we prove an analogue of this result in the setting of Hochschild homology of rings with anti-involution, where the relevant bar construction is the dihedral bar construction

Theorem 2.1 ([AGH21])**.** The dihedral bar construction satisfies the universal property of the norm $N_{C_2}^{O(2)}$.

One property of the norm for finite groups is the multiplicative double coset formula. For our new notion of norm $N_{D_2}^{O(2)}$, we also prove a such a formula, extending a result of [DMPP17] to Dihedral groups D_{2m} of order $2m$ for $m > 1$.

Theorem 2.2 ([AGH21])**.** There is an explicit muliplcative double coset formula for the norm $N_{D_2}^{O(2)}(R)$ restricted to D_{2m}.

We use these results to motivate a new algebraic definition of Real Hochschild homology $\underline{\mathrm{HR}}_*^{D_{2m}}(\underline{M})$ of a ring with anti-involution $\underline{M}$. We prove that this construction is equipped with structure maps R, F and V. and use this to define the p-typical Real Witt vectors $\underline{\mathbb{W}}(\underline{M}; p)$ and we compute $\underline{\mathbb{W}}(\underline{\mathbb{Z}}; p)$.

2.2.2 Homology of twisted G-rings

In joint work with Péroux and Merling [AMP], I extend the theory of norms to more some general compact Lie groups. To do this, I use the theory of crossed simplicial groups. The input of our construction is a twisted G-ring, defined as follows. Prerequistely, a group with parity is a group G with a group homomorphism $\varphi\colon G \to C_2 = \{-1, 1\}$.A twisted G-ring is then a ring with a G-action by abelian group homomorphisms where g acts by a ring homomorphism if $\varphi(g) = 1$ and a ring anti-homomorphism if $\varphi(g) = -1$.

Our construction also uses the theory of crossed simplicial groups developed by Z. Feidorowicz and J. L. Loday [FL91]. Crossed simplicial groups $\Delta\mathbf{G}$ are a common generalization of simplicial groups and Connes' cyclic category Λ. Given a crossed simplicial group $\Delta\mathbf{G}$, we construct a notion of Hochschild homology $\mathrm{TH}\mathbf{G}$ whose input is a twisted G_0-ring, where G_0 is a certain finite group associated to the crossed simplicial group $\Delta\mathbf{G}$.

In the study of twisted G-rings, we prove the following result, which we state as a pre-theorem since it is work in progress.

Pre-theorem 2.1. The category $\Delta\mathbf{G}_+$, given adjoining an initial object to the crossed simplicial group $\Delta\mathbf{G}$, encodes the structure of a twisted G-rings in the sense that a monoidal functor $\Delta\mathbf{G}_+ \to \mathsf{Sp}$ is the same data as a twisted G-ring.

Using this, we give a Nikolaus–Scholze-type construction of

$$N^{\mathbf{G}}_{\mathfrak{G}_0}(R) = \mathrm{TH}\mathbf{G}(R)$$

providing new combinatorial models for norm functors for certain compact Lie groups.

Of particular interest is the case of the quaternioic category $\Delta\mathbf{Q}$ whose associated Hochschild homology gives a construction of the norm

$$N^{\mathrm{Pin}(2)}_{C_4}(R).$$

We aim to use $N^{\mathrm{Pin}(2)}_{C_4}(R)$ to study involutive knot Floer homology in analogy with the relationship between Hochschild homology and knot Floer homology. As an example of an application we have the following goal.

Long-Term Goal 2.3. Construct a Hodge-to-de Rham spectral sequence for $N^{\mathrm{Pin}(2)}_{C_4}(R)$ that can be used to study equivariant knot Floer homology.

As an example computation, we compute $\mathrm{TH}\mathbf{G}(R[M])$ where R is a twisted G_0-ring and M is a twisted G_0-monoid, which we describe qualitatively here.

2.2.3 A deformations of C_p-equivariant homotopy

Just as in the case of the complex numbers discussed in Section 2.2, the category of Artin–Tate $\mathbb{R}$-motivic homotopy theory can be regarded as a deformation of Sp^{C_2} by [BHS20]. In particular, there is a Beti-realization functor

$$\mathrm{Be}_{C_2}\colon \mathrm{SH}^{\mathrm{AT}}(\mathbb{R}) \longrightarrow \mathsf{Sp}^{C_2}$$

that can be used to simplifiy the study of computations in C_2-equivariant stable homotopy theory. In the case of a cyclic group of order p where p is an odd prime, denoted C_p, there is not a known good replacement for $\mathrm{SH}(\mathbb{R})$. In work in progress with Behrens, Belmont, and Kong [ABBK], I aim to give a artificial construction of a category equipped with a realization functor to Sp^{C_p} using the theory of deformations of stable homotopy theory.

Deformations of homotopy theory were developed in work of [GIKR18] and [Pst18]. It was used in a key way by Burkland–Hahn–Senger [BHS20] to extend the cellular $\mathbb{R}$-motivic stable homotopy category in a way that allows for realization functors recovering more information about the homotopy groups of C_2-spectra.

Short-Term Goal 2.4. Construct a deformation of Borel C_p-spectra that recovers the a-complete Artin–Tate $\mathbb{R}$-motivic homotopy category at the prime $p = 2$.

At the moment, we have constructed a deformation of the Borel C_p-equivariant stable homotopy category. The key tools used to make this work is an adaptation of the classical modified Adams spectral sequence.

In future work, we plan construct a deformation of the full C_p-equivariant stable homotopy theory. To do this, we must construct deformations of each of the terms in the fracture square

$$\begin{tikzcd} \mathsf{Sp}^{C_p} \arrow[r] \arrow[d] & \mathsf{Sp}^{C_p}[a^{-1}] \arrow[d] \\ \left(\mathsf{Sp}^{C_p}\right)^{\wedge}_{a} \arrow[r] & \left(\mathsf{Sp}^{C_p}\right)^{\wedge}_{a}[a^{-1}]. \end{tikzcd}$$

This is already possible at $p = 2$ using work of Burkland–Hahn–Senger [BHS20], but at odd primes one does not have access to an analogue of the spectrum $MU_{\mathbb{R}}$ and therefore one must take a different approach.

Goal 2.5. Construct a deformation of C_p-spectra that recovers the full Artin–Tate $\mathbb{R}$-motivic homotopy category at the prime $p = 2$.

Ravenel has suggested that a nice C_p-equivariant Brown–Peterson spectrum BP could be used to resolve the remaining case of the odd primary Arf–Kervaire invariant problem in analogy with the solution of the 2-primary Kervaire invariant psroblem by [HHR16]. The key part of this being the existence of a nice Adams–Novikov spectral sequence based at this C_p-equivariant BP. Some progress in this direction appears in work of Hahn–Senger–Wilson [HSW20]. On hope is that one may skip the construction of BP and instead aim to construct an analogous spectral sequence in the setting of deformations of C_p-equivariant homotopy theory in order to resolve this conjecture.

Long-Term Goal 2.6. Resolve the 3-primary Arf–Kervaire invariant problem using an artificial 3-primary analogue of $\mathrm{SH}(\mathbb{R})$.

2.2.4 Syntomic cohomology for rings with anti-involution

In a completely different vein, one may attempt to carry out a thorough study of the arithmetic of equivariant ring spectra. For rings, there is still a lot that isn't known in

this context, for example one may want to define C_2-equivariant prismatic cohomology and syntomic cohomology. In joint work with J.D. Quigley [AQ], we propose to use the strong even filtration, a version of the even filtration of [HRW22] adapted to this context. We will call this filtration on Real topological Hochschild homology the motivic filtration in analogy with the filtration of [HRW22] which recovers the more classical motivic filtration in certain cases.

Goal 2.7. Compute $\mathrm{THR}(BP_{\mathbb{R}}\langle 1\rangle)$ using descent along the motivic spectral sequence.

Long term, we aim to understand syntomic/prismatic cohomology for suitable commutative monoids in C_2-spectra.

Long-Term Goal 2.8. Develop the theory of C_2-equivariant chromatically quasisyntomic genuine commutative C_2-rings and the notions of syntomic/prismatic cohomology.

References

[AAC+22] Gabriel Angelini-Knoll, Christian Ausoni, Dominic Leon Culver, Eva Höning, and John Rognes. Algebraic K-theory of elliptic cohomology. *arXiv e-prints*, page arXiv:2204.05890, April 2022.

[AAHR] Gabriel Angelini-Knoll, Christian Ausoni, Eva Höning, and John Rognes. G-theory of ring spectra and localization sequences. *In progress.*

[ABBK] Gabriel Angelini-Knoll, Mark Behrens, Eva Belmont, and Hana Kong. Artificial borel c_p-equivariant homotopy. *In progress.*

[ABG+18a] Vigleik Angeltveit, Andrew J. Blumberg, Teena Gerhardt, Michael A. Hill, Tyler Lawson, and Michael A. Mandell. Topological cyclic homology via the norm. *Doc. Math.*, 23:2101–2163, 2018.

[ABG18b] Benjamin Antieau, Tobias Barthel, and David Gepner. On localization sequences in the algebraic K-theory of ring spectra. *J. Eur. Math. Soc. (JEMS)*, 20(2):459–487, 2018.

[ACH] Gabriel Angelini-Knoll, Dominic Culver, and Eva Höning. Topological Hochschild homology of truncated Brown-Peterson spectra II. *In progress.*

[ACH21] Gabriel Angelini-Knoll, Dominic Leon Culver, and Eva Höning. Topological Hochschild homology of truncated Brown-Peterson spectra I. *arXiv e-prints*, page arXiv:2106.06785, June 2021.

[Ada58] J. F. Adams. On the structure and applications of the Steenrod algebra. *Comment. Math. Helv.*, 32:180–214, 1958.

[Ada66] J. F. Adams. On the groups $J(X)$. IV. *Topology*, 5:21–71, 1966.

[AGH21] Gabriel Angelini-Knoll, Teena Gerhardt, and Michael Hill. Real topological Hochschild homology via the norm and Real Witt vectors. *arXiv e-prints*, page arXiv:2111.06970, November 2021.

[AHL10] Vigleik Angeltveit, Michael A. Hill, and Tyler Lawson. Topological Hochschild homology of ℓ and *ko*. *Amer. J. Math.*, 132(2):297–330, 2010.

[AHW] Gabriel Angelini-Knoll, Jeremy Hahn, and Dylan Wilson. Syntomic cohomology and K-theory of Morava K-theory. *In progress*.

[AK17] Gabriel J. Angelini-Knoll. *Periodicity in Iterated Algebraic K-Theory of Finite Fields*. ProQuest LLC, Ann Arbor, MI, 2017. Thesis (Ph.D.)–Wayne State University.

[AK18] Gabriel Angelini-Knoll. Detecting β elements in iterated algebraic K-theory of finite fields. *arXiv e-prints*, page arXiv:1810.10088, October 2018.

[AK21] Gabriel Angelini-Knoll. On topological Hochschild homology of the $K(1)$-local sphere. *Journal of Topology*, 14(1):258–290, 2021.

[AK22] Gabriel Angelini-Knoll. Complex orientations and TP of complete DVRS. *Accepted in Homol. Homotopy Appl.*, 2022.

[AMP] Gabriel Angelini-Knoll, Mona Merling, and Maximilien Péroux. Homology of twisted g-rings. *In progress*.

[AQ] Gabriel Angelini-Knoll and J.D. Quigley. Equivariant motivic filtrations. *In progress*.

[AQ19] Gabriel Angelini-Knoll and J. D. Quigley. Chromatic Complexity of the Algebraic K-theory of $y(n)$. *arXiv e-prints*, page arXiv:1908.09164, August 2019.

[AQ21] Gabriel Angelini-Knoll and J. D. Quigley. The Segal conjecture for topological Hochschild homology of Ravenel spectra. *Journal of Homotopy and Related Structures*, 2021.

[AR02] Christian Ausoni and John Rognes. Algebraic K-theory of topological K-theory. *Acta Math.*, 188(1):1–39, 2002.

[AR08] Christian Ausoni and John Rognes. *The chromatic red-shift in algebraic K-theory*, volume 40 of *Monogr. Enseign. Math.* L'Enseignement Mathématique, Geneva, 2008.

[AR12] Christian Ausoni and John Rognes. Algebraic K-theory of the first Morava K-theory. *J. Eur. Math. Soc. (JEMS)*, 14(4):1041–1079, 2012.

[AS18] Gabe Angelini-Knoll and Andrew Salch. A May-type spectral sequence for higher topological Hochschild homology. *Algebr. Geom. Topol.*, 18(5):2593–2660, 2018. agt.2018.18.2593.

[AS20] Gabriel Angelini-Knoll and Andrew Salch. Commuting unbounded homotopy limits with Morava K-theory. *Submitted.*, 2020. arXiv:2003.03510.

[Beh06] Mark Behrens. A modular description of the $K(2)$-local sphere at the prime 3. *Topology*, 45(2):343–402, 2006.

[BHM93] M. Bökstedt, W. C. Hsiang, and I. Madsen. The cyclotomic trace and algebraic K-theory of spaces. *Invent. Math.*, 111(3):465–539, 1993.

[BHS19] Robert Burklund, Jeremy Hahn, and Andrew Senger. On the boundaries of highly connected, almost closed manifolds. *arXiv e-prints*, page arXiv:1910.14116, October 2019.

[BHS20] Robert Burklund, Jeremy Hahn, and Andrew Senger. Galois reconstruction of Artin-Tate $\mathbb{R}$-motivic spectra. *arXiv e-prints*, page arXiv:2010.10325, October 2020.

[BL09] Mark Behrens and Gerd Laures. β-family congruences and the f-invariant. In *New topological contexts for Galois theory and algebraic geometry (BIRS 2008)*, volume 16 of *Geom. Topol. Monogr.*, pages 9–29. Geom. Topol. Publ., Coventry, 2009.

[BL22] Bhargav Bhatt and Jacob Lurie. Absolute prismatic cohomology. *arXiv e-prints*, page arXiv:2201.06120, January 2022.

[BM08] Andrew J. Blumberg and Michael A. Mandell. The localization sequence for the algebraic K-theory of topological K-theory. *Acta Math.*, 200(2):155–179, 2008.

[BMS19] Bhargav Bhatt, Matthew Morrow, and Peter Scholze. Topological Hochschild homology and integral p-adic Hodge theory. *Publ. Math. Inst. Hautes Études Sci.*, 129:199–310, 2019.

[Bök87] M. Bökstedt. The topological Hochschild homology of $\mathbb{Z}$ and of $\mathbb{Z}/p\mathbb{Z}$. preprint, 1987.

[BS19] Bhargav Bhatt and Peter Scholze. Prisms and Prismatic Cohomology. *arXiv e-prints*, page arXiv:1905.08229, May 2019.

[BSY22] Robert Burklund, Tomer M. Schlank, and Allen Yuan. The Chromatic Nullstellensatz. *arXiv e-prints*, page arXiv:2207.09929, July 2022.

[CMNN20] Dustin Clausen, Akhil Mathew, Niko Naumann, and Justin Noel. Descent in algebraic K-theory and a conjecture of Ausoni-Rognes. *J. Eur. Math. Soc. (JEMS)*, 22(4):1149–1200, 2020.

[DGM13] Bjørn Ian Dundas, Thomas G. Goodwillie, and Randy McCarthy. *The local structure of algebraic K-theory*, volume 18 of *Algebra and Applications*. Springer-Verlag London, Ltd., London, 2013.

[DMPP17] E. Dotto, K. Moi, I. Patchkoria, and S. Precht Reeh. Real topological Hochschild homology. *ArXiv e-prints*, November 2017.

[FL91] Zbigniew Fiedorowicz and Jean-Louis Loday. Crossed simplicial groups and their associated homology. *Trans. Amer. Math. Soc.*, 326(1):57–87, 1991.

[Fre38] Hans Freudenthal. über die Klassen der Sphärenabbildungen I. Große Dimensionen, 1938.

[GIKR18] Bogdan Gheorghe, Daniel C. Isaksen, Achim Krause, and Nicolas Ricka. C-motivic modular forms. 10 2018.

[Hah16] Jeremy Hahn. On the Bousfield classes of H_∞-ring spectra. *arXiv e-prints*, page arXiv:1612.04386, December 2016.

[HHR16] M. A. Hill, M. J. Hopkins, and D. C. Ravenel. On the nonexistence of elements of Kervaire invariant one. *Ann. of Math. (2)*, 184(1):1–262, 2016.

[HL10] Michael Hill and Tyler Lawson. Automorphic forms and cohomology theories on Shimura curves of small discriminant. *Adv. Math.*, 225(2):1013–1045, 2010.

[HM03] Lars Hesselholt and Ib Madsen. On the K-theory of local fields. *Ann. of Math. (2)*, 158(1):1–113, 2003.

[HN19] Lars Hesselholt and Thomas Nikolaus. *Topological cyclic homology*. CRC Press, 2019.

[HNS22] Yonaton Harpaz, Thomas Nikolaus, and J. Shah. The real cyclotomic trace map. *In progress*, 2022.

[Hön21] Eva Höning. The topological Hochschild homology of algebraic K-theory of finite fields. *Ann. K-Theory*, 6(1):29–96, 2021.

[HRW22] Jeremy Hahn, Arpon Raksit, and Dylan Wilson. A motivic filtration on the topological cyclic homology of commutative ring spectra. *arXiv e-prints*, page arXiv:2206.11208, June 2022.

[HSW20] Jeremy Hahn, Andrew Senger, and Dylan Wilson. Odd primary analogs of Real orientations. *arXiv e-prints*, page arXiv:2009.12716, September 2020.

[HW20] Jeremy Hahn and Dylan Wilson. Redshift and multiplication for truncated Brown-Peterson spectra. *arXiv e-prints*, page arXiv:2012.00864, December 2020.

[IWX20] Daniel C. Isaksen, Guozhen Wang, and Zhouli Xu. Stable homotopy groups of spheres. *Proceedings of the National Academy of Sciences*, 117(40):24757–24763, 2020.

[Lau99] Gerd Laures. The topological q-expansion principle. *Topology*, 38(2):387–425, 1999.

[Law18] Tyler Lawson. Secondary power operations and the Brown-Peterson spectrum at the prime 2. *Ann. Math. (2)*, 188(2):513–576, 2018.

[Lic73] Stephen Lichtenbaum. Values of zeta-functions, étale cohomology, and algebraic K-theory. pages 489–501. Lecture Notes in Math., Vol. 342, 1973.

[LMMT20] Markus Land, Akhil Mathew, Lennart Meier, and Georg Tamme. Purity in chromatically localized algebraic K-theory. *arXiv e-prints*, page arXiv:2001.10425, January 2020.

[LN12] Tyler Lawson and Niko Naumann. Commutativity conditions for truncated Brown-Peterson spectra of height 2. *J. Topol.*, 5(1):137–168, 2012.

[Mah79] Mark Mahowald. Ring spectra which are Thom complexes. *Duke Math. J.*, 46(3):549–559, 1979.

[MR99] Mark Mahowald and Charles Rezk. Brown-Comenetz duality and the Adams spectral sequence. *Amer. J. Math.*, 121(6):1153–1177, 1999.

[MRS01] Mark Mahowald, Douglas Ravenel, and Paul Shick. The triple loop space approach to the telescope conjecture. In *Homotopy methods in algebraic topology (Boulder, CO, 1999)*, volume 271 of *Contemp. Math.*, pages 217–284. Amer. Math. Soc., Providence, RI, 2001.

[MRW75] Haynes R. Miller, Douglas C. Ravenel, and W. Stephen Wilson. Novikov's Ext^2 and the nontriviality of the gamma family. *Bull. Am. Math. Soc.*, 81:1073–1075, 1975.

[NS18] Thomas Nikolaus and Peter Scholze. On topological cyclic homology. *Acta Math.*, 221(2):203–409, 2018.

[Pst18] Piotr Pstragowski. Synthetic spectra and the cellular motivic category. 03 2018.

[QS21] J. D. Quigley and Jay Shah. On the equivalence of two theories of real cyclotomic spectra. *arXiv e-prints*, page arXiv:2112.07462, December 2021.

[Qui69] Daniel Quillen. On the formal group laws of unoriented and complex cobordism theory. *Bull. Amer. Math. Soc.*, 75:1293–1298, 1969.

[Qui72] Daniel Quillen. On the cohomology and K-theory of the general linear groups over a finite field. *Ann. of Math. (2)*, 96:552–586, 1972.

[Qui73] Daniel Quillen. Higher algebraic K-theory. I. In *Algebraic K-theory, I: Higher K-theories (Proc. Conf., Battelle Memorial Inst., Seattle, Wash., 1972)*, pages 85–147. Lecture Notes in Math., Vol. 341. Springer, Berlin, 1973.

[Qui75] Daniel Quillen. Higher algebraic K-theory. In *Proceedings of the International Congress of Mathematicians (Vancouver, B. C., 1974), Vol. 1*, pages 171–176. Canad. Math. Congress, Montreal, Que., 1975.

[Rog00] John Rognes. Algebraic k-theory of

finitely presented ring spectra, 2000. Talk at Oberwolfach.

[Rog14] John Rognes. Algebraic K-theory of strict ring spectra. In *Proceedings of the International Congress of Mathematicians—Seoul 2014. Vol. II*, pages 1259–1283. Kyung Moon Sa, Seoul, 2014.

[Sen17] Andrew Senger. The Brown-Peterson spectrum is not $\mathbb{E}_{2(p^2+2)}$ at odd primes. *arXiv e-prints*, page arXiv:1710.09822, October 2017.

[Voe11] Vladimir Voevodsky. On motivic cohomology with $\mathbf{Z}/l$-coefficients. *Ann. of Math. (2)*, 174(1):401–438, 2011.

[Yua21] Allen Yuan. Examples of chromatic redshift in algebraic K-theory. *arXiv e-prints*, page arXiv:2111.10837, November 2021.